

Quine McKluskey (Table Method)

Quine-McKluskey Method
(Table Method)

$$Y(a,b,c) = \sum m(0,1,2,3,6)$$

bc	00	01	11	10
a				
0	1	1	1	1
1				1

$$y = \bar{a} + b\bar{c}$$

TABLE METHOD

1) Step 1: Write binary values of given min terms

$$0 = 000$$

$$1 = 001$$

$$2 = 010$$

$$3 = 011$$

$$6 = 110$$

Step 2: Group the miniterms based on the number of '1' bits in each miniterm

miniterm	no. of 1s	Group
0 (000)	0	Group 0 (G_0)
1 (001)	1	Group 1 (G_1)
2 (010)	1	
3 (011)	2	Group 2 (G_2)
6 (110)	2	

Step 3: Find pairs from adjacent groups

$$\begin{aligned}
 &\cancel{G_0, G_1: (0, 1) =} \\
 &\checkmark G_0, G_1: (m_0, m_1) = 00- \\
 &\checkmark G_0, G_2: (m_0, m_2) = 0-0 \\
 &\checkmark G_1, G_2: (m_1, m_3) = 0-1 \\
 &\times \quad \quad \quad (m_1, m_6) \times \\
 &\checkmark \quad \quad \quad (m_2, m_3) \times = 01- \\
 &\quad \quad \quad (m_2, m_6) = -10
 \end{aligned}$$

Step 3: Find quads using the pairs in step 3

$$\begin{aligned} (m_0, m_2), (m_1, m_3) &= 0-- \\ \text{drop} \rightarrow (m_0, m_1), (m_2, m_3) &= 0-- \\ (\text{repeated}) \end{aligned}$$

$$Y = (m_2, m_1) + (m_0, m_2, m_1, m_3)$$

$$= -10 + 0--$$

$$= b\bar{c} + \bar{a}$$

$$Y = \sum m(0, 3, 7) \rightarrow \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{array}$$

~~$$G_0 = 000$$~~

$$G_0 \mid 0 \mid 000$$

$$G_1 \mid - \mid$$

$$G_2 \mid 3 \mid 011$$

$$G_3 \mid 7 \mid 111$$

$$G_0 = 000$$

$$(3, 7) = -11$$

$$\therefore Y = \bar{a}\bar{b}\bar{c} + bc$$

		bc			
		00	01	11	10
a	0	1		1	
	1			1	

$$y = \sum m(0, 2, 4, 6)$$

$$y = \sum m(0, 1, 8, 9, 11, 13, 15)$$

ab \ cd	00	01	11	10
00	1	1		
01				
11		1	1	
10	1	1	1	

$$y = \bar{b}\bar{c} + \bar{a}d$$

Table method

Step 1

- ✓ 0 = 0000 — G0
- ✓ 1 = 0001 ↗ G1
- ✓ 8 = 1000 ↘ G1
- ✓ 9 = 1001 ← G2
- ✓ 11 = 1011 → G3
- ✓ 13 = 1101 → G3
- ✓ 15 = 1111 — G4

Step 2

G₀, G₁

✓ (0, 8) = 000 -

✓ (0, 8) = -000

G₁, G₂

✓ (1, 9) = -001

✓ (8, 9) = ~~100~~ -

G₂, G₃

✓ (9, 11) = 10 -

✓ (9, 13) = 1 - 01

G₃, G₄

✓ (11, 15) = 1 - 11

✓ (13, 15) = 11 - 1

Step 3 Find quads

(0, 1, 8, 9) = -00 -

(0, 8, 1, 9) = -00 - ← repeated

(9, 11, 13, 15) = 1 - - 1

(9, 13, 11, 15) = 1 - - 1 ← repeated

Step 4

$$y = -00 - + 1 - - 1$$

$$= \overline{b}\overline{c} + ad$$